Exercise 1. Suppose $\{x_n\}$ is a sequence and suppose for some $x \in \mathbb{R}$, the limit

$$\lim_{n\to\infty} \frac{|x_{n+1} - x|}{|x_n - x|} < 1.$$

Show that $\{x_n\}$ converges to x.

Proof. Let $\{x_n\}$ be any sequence of $\mathbb{R}$. Suppose $x \in \mathbb{R}$ and

$$0 < \lim_{n\to\infty} \frac{|x_{n+1} - x|}{|x_n - x|} < 1.$$

Choose $\alpha \in (0, 1)$ such that

$$\lim_{n\to\infty} \frac{|x_{n+1} - x|}{|x_n - x|} < \alpha.$$

Then $\exists M \in \mathbb{N}$ for which

$$\forall n \geq M : \frac{|x_{n+1} - x|}{|x_n - x|} \leq \alpha \iff |x_{n+1} - x| \leq \alpha |x_n - x|$$

and

$$\forall n \geq M : |x_{n+m} - x| \leq \alpha^m |x_n - x| \implies |x_{n+m+1} - x| \leq \alpha |x_{n+m} - x| \leq \alpha^{m+1} |x_n - x|;$$

hence,

$$\forall n \geq M : 0 \leq |x_n - x| \leq \alpha^{n-M} |x_M - x|.$$

By the Squeeze Theorem of $\mathbb{R}$,

$$\forall n \geq M : 0 \leftarrow 0 \leq |x_n - x| \leq \alpha^{n-M} |x_M - x| \to 0 \implies \lim_{n\to\infty} |x_n - x| = 0.$$

Thus, $\{x_n\}$ converges to x, as needed. □

Exercise 2.

a) Let $x_n := (-1)^n/n$. Find $\limsup x_n$ and $\liminf x_n$.

b) Let $x_n := (n-1)(-1)^n/n$. Find $\limsup x_n$ and $\liminf x_n$.

Proof. Start!

a) Let

$$\forall n \in \mathbb{N},\ x_n := \frac{(-1)^n}{n} \in \mathbb{R}.$$

Then

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup \left\{ \frac{(-1)^k}{k} : k \geq n \right\} = \lim_{n\to\infty} \frac{1}{2n} = 0$$

and

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf \left\{ \frac{(-1)^k}{k} : k \geq n \right\} = \lim_{n\to\infty} \frac{1}{1-2n} = 0.$$

b) Let

$$\forall n \in \mathbb{N},\ x_n := \frac{(n-1)(-1)^n}{n} = (-1)^n \left(1 - \frac{1}{n}\right) \in \mathbb{R}.$$

Then

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\left\{(-1)^k \left(1 - \frac{1}{k}\right) : k \geq n\right\} = \lim_{n\to\infty} 1 = 1$$

and

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\left\{(-1)^k \left(1 - \frac{1}{k}\right) : k \geq n\right\} = \lim_{n\to\infty} -1 = -1.$$

□

Exercise 3. Let $\{x_n\}$ and $\{y_n\}$ be bounded sequences such that $x_n \leq y_n$ for all n. Then show that

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n$$

and

$$\liminf_{n\to\infty} x_n \leq \liminf_{n\to\infty} y_n.$$

Proof. Let $\{x_n\}$ and $\{y_n\}$ be any bounded sequences of $\mathbb{R}$. Suppose $x_n \leq y_n$ for all $n \in \mathbb{N}$. If $\sup\{x_k \in \mathbb{R} : k \geq n\} > \sup\{y_k \in \mathbb{R} : k \geq n\}$ for any $n \in \mathbb{N}$, then $\exists N \in \mathbb{N}$ such that $\sup\{x_k : k \geq n\} \geq x_N > \sup\{y_n : k \geq n\} \geq y_N$; hence, $x_N > y_N$ for some $N \in \mathbb{N}$ and $\sup\{x_k \in \mathbb{R} : k \geq n\} \leq \sup\{y_k \in \mathbb{R} : k \geq n\}$ for all $n \in \mathbb{N}$. Thus,

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{x_k \in \mathbb{R} : k \geq n\} \leq \lim_{n\to\infty} \sup\{y_k \in \mathbb{R} : k \geq n\} = \limsup_{n\to\infty} y_n,$$

as needed. If $\inf\{x_k \in \mathbb{R} : k \geq n\} > \inf\{y_k \in \mathbb{R} : k \geq n\}$ for any $n \in \mathbb{N}$, then $\exists N \in \mathbb{N}$ such that $x_N \geq \inf\{x_k : k \geq n\} > y_N \geq \inf\{y_k : k \geq n\}$; hence, $x_N > y_N$ for some $N \in \mathbb{N}$ and $\inf\{x_k \in \mathbb{R} : k \geq n\} \leq \inf\{y_k \in \mathbb{R} : k \geq n\}$ for all $n \in \mathbb{N}$. Thus,

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\{x_k \in \mathbb{R} : k \geq n\} \leq \lim_{n\to\infty} \inf\{y_k \in \mathbb{R} : k \geq n\} = \liminf_{n\to\infty} y_n,$$

as needed. □

Exercise 4. Let $\{x_n\}$ and $\{y_n\}$ be bounded sequences.

a) Show that $\{x_n + y_n\}$ is bounded.

b) Show that

$$\liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n \leq \liminf_{n\to\infty} (x_n + y_n).$$

Hint: Find a subsequence $\{x_{n_i} + y_{n_i}\}$ of $\{x_n + y_n\}$ that converges. Then find a subsequence $\{x_{n_{m_i}}\}$ of $\{x_{n_i}\}$ that converges. Then apply what you know about limits.

c) Show that

$$\limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n \geq \limsup_{n\to\infty} (x_n + y_n).$$

Hint: See previous hint.

d) Find an explicit $\{x_n\}$ and $\{y_n\}$ such that

$$\liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n < \liminf_{n\to\infty}(x_n + y_n).$$

Hint: Look for examples that do not have a limit.

e) Find an explicit $\{x_n\}$ and $\{y_n\}$ such that

$$\limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n > \limsup_{n\to\infty}(x_n + y_n).$$

Hint: See previous hint.

Proof. Let $\{x_n\}$ and $\{y_n\}$ be any bounded sequences of $\mathbb{R}$.

a) By the Least Upper Bound Property of $\mathbb{R}$,

$$\forall n \in \mathbb{N} : \inf_{k\in\mathbb{N}} x_k \leq x_n \leq \sup_{k\in\mathbb{N}} x_k$$

and

$$\forall n \in \mathbb{N} : \inf_{k\in\mathbb{N}} y_k \leq y_n \leq \sup_{k\in\mathbb{N}} y_k.$$

Thus,

$$\forall n \in \mathbb{N} : \inf_{k\in\mathbb{N}} x_k + \inf_{k\in\mathbb{N}} y_k \leq x_n + y_n \leq \sup_{k\in\mathbb{N}} x_k + \sup_{k\in\mathbb{N}} y_k,$$

as needed.

b) By the Bolzano-Weierstrass Theorem of $\mathbb{R}$, $\exists$ convergent subsequence $\{x_{n_i} + y_{n_i}\}$ of the sequence $\{x_n + y_n\}$ such that

$$\lim_{i\to\infty}(x_{n_i} + y_{n_i}) = \liminf_{n\to\infty}(x_n + y_n)$$

and $\exists$ convergent subsequence $\{x_{n_{m_i}}\}$ of the sequence $\{x_{n_i}\}$ such that

$$\lim_{i\to\infty} x_{n_{m_i}} = \limsup_{n\to\infty} x_n.$$

Then $\{x_{n_{m_i}} + y_{n_{m_i}}\}$ is a convergent subsequence of the convergent sequence $\{x_{n_i} + y_{n_i}\}$ for which

$$\lim_{i\to\infty}(x_{n_{m_i}} + y_{n_{m_i}}) = \lim_{i\to\infty}(x_{n_i} + y_{n_i}).$$

By the Limit Laws of $\mathbb{R}$, $\{y_{n_{m_i}}\}$ is a convergent subsequence of the sequence $\{y_{n_i}\}$ such that

$$\lim_{i\to\infty} y_{n_{m_i}} = \lim_{i\to\infty}(x_{n_{m_i}} + y_{n_{m_i}} - x_{n_{m_i}}) = \lim_{i\to\infty}(x_{n_{m_i}} + y_{n_{m_i}}) - \lim_{i\to\infty} x_{n_{m_i}}.$$

Thus, $\{x_n + y_n\}$ is a sequence of $\mathbb{R}$ for which

$$\begin{aligned}
\liminf_{n\to\infty}(x_n + y_n) &= \lim_{i\to\infty}(x_{n_i} + y_{n_i}) \\
&= \lim_{i\to\infty}(x_{n_{m_i}} + y_{n_{m_i}}) \\
&= \lim_{i\to\infty} x_{n_{m_i}} + \lim_{i\to\infty} y_{n_{m_i}} \\
&= \limsup_{n\to\infty} x_n + \lim_{i\to\infty} y_{n_{m_i}} \geq \liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n,
\end{aligned}$$

as needed.

c) By the Bolzano-Weierstrass Theorem of $\mathbb{R}$, $\exists$ convergent subsequence $\{x_{n_j} + y_{n_j}\}$ of the sequence $\{x_n + y_n\}$ such that

$$\lim_{j\to\infty}(x_{n_j} + y_{n_j}) = \limsup_{n\to\infty}(x_n + y_n)$$

and $\exists$ convergent subsequence $\{y_{n_{m_j}}\}$ of the sequence $\{y_{n_j}\}$ such that

$$\lim_{j\to\infty} y_{n_{m_j}} = \liminf_{n\to\infty} y_n.$$

Then $\{x_{n_{m_j}} + y_{n_{m_j}}\}$ is a convergent subsequence of the convergent sequence $\{x_{n_j} + y_{n_j}\}$ for which

$$\lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) = \lim_{j\to\infty}(x_{n_j} + y_{n_j}).$$

By the Limit Laws of $\mathbb{R}$, $\{x_{n_{m_j}}\}$ is a convergent subsequence of the sequence $\{x_{n_j}\}$ such that

$$\lim_{j\to\infty} x_{n_{m_j}} = \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}} - y_{n_{m_j}}) = \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) - \lim_{j\to\infty} y_{n_{m_j}}.$$

Thus, $\{x_n + y_n\}$ is a sequence of $\mathbb{R}$ for which

$$\begin{aligned}\limsup_{n\to\infty}(x_n + y_n) &= \lim_{j\to\infty}(x_{n_j} + y_{n_j}) \\ &= \lim_{j\to\infty}(x_{n_{m_j}} + y_{n_{m_j}}) \\ &= \lim_{j\to\infty} x_{n_{m_j}} + \lim_{j\to\infty} y_{n_{m_j}} \\ &= \lim_{j\to\infty} x_{n_{m_j}} + \liminf_{n\to\infty} y_n \le \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n,\end{aligned}$$

as needed.

d) If $x_n = (-1)^n$ and $y_n = (-1)^{n+1}$, then

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\{(-1)^k : k \ge n\} = \lim_{n\to\infty} -1 = -1$$

and

$$\liminf_{n\to\infty} y_n = \lim_{n\to\infty} \inf\{(-1)^{k+1} : k \ge n\} = \lim_{n\to\infty} -1 = -1$$

and

$$\liminf_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} \inf\{(-1)^k + (-1)^{k+1} : k \ge n\} = \lim_{n\to\infty} 0 = 0;$$

hence,

$$\liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n = -2 < 0 = \liminf_{n\to\infty}(x_n + y_n),$$

as needed.

e) If $x_n = (-1)^n$ and $y_n = (-1)^{n+1}$, then

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{(-1)^k : k \ge n\} = \lim_{n\to\infty} 1 = 1$$

and

$$\limsup_{n\to\infty} y_n = \lim_{n\to\infty} \sup\{(-1)^{k+1} : k \geq n\} = \lim_{n\to\infty} 1 = 1$$

and

$$\limsup_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} \sup\{(-1)^k + (-1)^{k+1} : k \geq n\} = \lim_{n\to\infty} 0 = 0;$$

hence,

$$\limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n = 2 > 0 = \limsup_{n\to\infty}(x_n + y_n),$$

as needed.

□

Exercise 5. Let $\{x_n\}$ be a bounded sequence of real numbers. Prove

$$\lim_{n\to\infty} x_n = 0 \iff \limsup_{n\to\infty} |x_n| = 0.$$

Proof. Let $\{x_n\}$ be any bounded sequence of $\mathbb{R}$. Then

$$\forall n \in \mathbb{N} : |x_n| \leq \sup\{|x_k| \in \mathbb{R} \mid k \geq n\}.$$

We want to prove directly that

$$\lim_{n\to\infty} x_n = 0 \implies \limsup_{n\to\infty} |x_n|.$$

By the Limit Characterization Theorem,

$$\lim_{n\to\infty} x_n = 0 \implies \lim_{n\to\infty} |x_n| = 0 \implies \limsup_{n\to\infty} |x_n| = 0.$$

We want to prove conversely that

$$\limsup_{n\to\infty} |x_n| = 0 \implies \lim_{n\to\infty} x_n = 0.$$

By the Squeeze Theorem,

$$\limsup_{n\to\infty} |x_n| = 0 \implies |x_n| \leq \sup\{x_k \in \mathbb{R} \mid k \geq n\} \to 0 \implies \lim_{n\to\infty} x_n = 0.$$

□

Exercise 6. Does there exist a sequence $\{x_n\}$ such that

$$\liminf_{n\to\infty} x_n = -1, \ \lim_{n\to\infty} x_n = 0, \ \limsup_{n\to\infty} x_n = 1?$$

Either give an example, or explain why no such example exists.

Proof. No real example exists! If $\{x_n\}$ is any sequence of $\mathbb{R}$, then

$$\lim_{n\to\infty} x_n = 0 \iff \liminf_{n\to\infty} = \limsup_{n\to\infty} = 0;$$

hence,

$$\liminf_{n\to\infty} x_n = -1 \implies \lim_{n\to\infty} x_n \neq 0$$

and

$$\lim_{n\to\infty} x_n = 0 \implies \limsup_{n\to\infty} x_n \neq 1,$$

as needed. □

Exercise 7. True or false, prove or find a counterexample: If $\{x_n\}$ is a Cauchy sequence, then there exists an M such that for all $n \geq M$ we have $|x_{n+1} - x_n| \leq |x_n - x_{n-1}|$.

Proof. Let

$$\forall k \in \mathbb{N} : x_{2k-1} = 18.100 + \sum_{j=1}^{k-1} \frac{(-1)^j}{j^2} + \sum_{j=1}^{k-1} \frac{(-1)^j}{2^j} \in \mathbb{R}$$

and

$$\forall k \in \mathbb{N} : x_{2k} = 18.100 + \sum_{j=1}^{k-1} \frac{(-1)^j}{j^2} + \sum_{j=1}^{k} \frac{(-1)^j}{2^j} \in \mathbb{R}.$$

Then

$$\forall k \in \mathbb{N} : |x_{2k} - x_{2k-1}| = \frac{1}{2^k} \to 0$$

and

$$\forall k \in \mathbb{N} : |x_{2k+1} - x_{2k}| = \frac{1}{k^2} \to 0.$$

Thus,

$$\forall \epsilon > 0 \; \exists M \in \mathbb{N} \; \forall m, n \geq M : |x_m - x_n| < \epsilon$$

and

$$\forall M \in \mathbb{N} \; \exists n \geq M : |x_{n+1} - x_n| > |x_n - x_{n-1}|,$$

as needed. □